Nurture vs. Nature: The Effects of Location and Experience on Hurricane Adaptation*

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Abstract

Natural disasters have become more frequent and severe causing some people to experience them multiple times in quick succession. Additionally, changes in weather patterns and atmospheric conditions have made previously predictable aspects of natural disasters less so. While increasing experience with disasters may give people opportunities to learn how to prepare, any long lapses between subsequent disasters may result in discounting of the disaster's severity. I study the effects of experience and discounting on the average response to natural disasters by analyzing sales of emergency supplies, namely bottle water, during the 2008-2019 Atlantic hurricane seasons. Through event study analysis, I find that increases in historical exposure have little effect on the responsiveness of consumers; however, consumers respond increasingly to current hurricanes as the length of time since the most recent hurricane increases.

*I thank John Chalmers at the University of Oregon for providing access to the NielsenIQ Datasets. This paper represents the researcher's own analyses calculated (or derived) based in part on data from Nielsen Consumer LLC and marketing databases provided through the NielsenIQ Datasets at the Kilts Center for Marketing Data Center at The University of Chicago Booth School of Business. Conclusions drawn from the NielsenIQ data are those of the researcher and do not reflect the views of NielsenIQ. NielsenIQ is not responsible for, had no role in, and was not involved in analyzing and preparing the results reported herein

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1 Introduction

Natural disasters are becoming more frequent and severe (Anderson and Bausch, 2006; Van Aalst, 2006). This trend has led to broad impacts on the economy as natural disasters cause poorer labor market outcomes, decreased land values, stalls in international trade, and increased government transfer payments (Ibarrarán et al., 2009; Deryugina, 2017; Deryugina et al., 2018; Sytsma, 2020). As disasters become more frequent, individuals living in threatened areas may experience an increasing number of disasters during their lifespan. This frequency allows people more opportunities to learn from disasters and use that information when preparing for future disasters. However, as storm patterns change, people who live in areas which previously experienced few storms may face disasters without having had prior learning opportunities. This lack of experience and information could lead to people not properly preparing when a disaster finally affects them. Additionally, because many years can pass before a person faces a subsequent disaster, the average person may face a discounting problem. If significant time has passed, the people may discount the effect of disasters and by extension, their need to remain prepared. The destruction caused by natural disasters can pose difficulties and safety hazards for people purchasing emergency supplies after the disaster. Hazardous road conditions after natural disasters can also make it more difficult for relief efforts to provide support for those who were not able to properly prepare (Horner and Widener, 2011). In some areas, a lack of experience with storms or extreme discounting due to a lack of recent disasters may lead to an increase in the required funds and efforts needed to restore the area. Thus, it is important to understand how experience and discounting of disasters influences preparation in the future. By identifying any gaps in responsiveness and the mechanisms that drive them, firms can better prepare for the response of their customers and avoid any shortages or supply chain issues that may arise. Understanding the mechanisms driving the response to natural disasters would also allow government agencies to make and propose policies that target the issues before hand, rather than waiting to respond after the fact.

In this paper, I explore how experience with tropical disturbances and the time between subsequent disturbances affects purchasing patterns of emergency supplies in the Atlantic and Gulf

Coast regions of the United States (i.e. areas most often subject to the annual Atlantic hurricane season). To understand this relationship, I combine NielsenIQ Scanner Data with geographic forecast data from the National Oceanic and Atmospheric Association (NOAA) from 2008-2019. With modern forecasting techniques, consumers can respond to the threat of a hurricane days before it lands.[1] This allows them to purchase emergency supplies in response to the forecast rather than the hurricane itself. Using the scanner data I observe the sale of emergency supplies throughout the year for nearly all counties in the Southeast. Additionally, I employ NOAA's historical hurricane landfall data (HURDAT2) in order to build out each county's hurricane history.

My analysis begins with a baseline exploration of the effect of hurricane forecasts and landfalls on the sales of different emergency supplies in the style of Beatty et al. (2019), before deviating to explore the effects of historical county attributes. I focus solely on sales of purified drinking water. While there are many items on the Department of Homeland Security's emergency kit checklist, I believe water most easily captures the responsiveness of the population. The reasons for this choice are that water is the first item on the list, it is the most homogeneous product on the list, it is readily available at nearly every store nation-wide, and it is often in the news and social media during natural disasters. I find that sales of bottled water increase from 20-30% during the weeks surrounding a hurricane.

I then diverge from their analysis by exploring the effects of county specific historical tropical disturbance variables on sales of emergency supplies during a current tropical disturbance. The first historical variable I analyze is the county's total historical landfall count since 1983. This allows me to test for the possibility of learning behavior by understanding how the sales in counties change as the exposure to tropical disturbances increases[2]. I find that for each additional hurricane a county previously faced, sales of batteries increase by 2.67% and sales of flashlights increase by 5.37% when the county faces a current tropical disturbance. Sales of bottled water at this time also increase,

[1] Hurricane seasons produce cyclonic storms of different intensities, including storms labeled by NOAA as "disturbances", "tropical depressions", "tropical storms", and "hurricanes", all of which can cause damage to infrastructure and housing. For simplicity I refer to all these categories of weather events as "hurricanes" throughout this paper.

[2] While this test can reject learning behavior, it is likely limited to confirm learning behavior, in part because I do not observe individual-level exposure. Additionally, migration between counties would imply that the average resident will have less exposure than the county itself which would reduce power and would skew the results to reject learning even if it exists. I assume that the results are representative of an average consumer that has lived in the county during the 12 years for which I have scanner data.

but the change is insignificant.

The second historical variable, years since the last storm, allows me to test for signs of salience of the disasters risk. If a county has not been hit for many years, regardless of its historical exposure, people's perceptions of the risk may start to fade. I find that sales of emergency supplies are significantly higher when storm follows a tropical storm or hurricane within the next six years. However, if a storm follows a major hurricane the largest increase in sales is only within the next 3 years. I also run a regression to test the effect of a single additional year since the last hurricane and find that sales of water increase by 1.34% and sales of flashlights increase by 2.68% when a county is in the predicted path of a later storm. Sales of batteries increase during this time, but the change is not significant.

Much of the current literature on natural disasters focuses on changes in markets after a disaster. Such literature includes effects of insurance take-up rates, labor market impacts, and government assistance applications (Hallstrom and Smith, 2005; Gallagher, 2014; McCoy and Walsh, 2018; Bakkensen et al., 2019; Groen and Polivka, 2008; Deryugina, 2017; Deryugina et al., 2018). Others focus on how markets and decisions are made in the days leading up to a singular disaster through the use of phone surveys (Kelly et al., 2012; Meyer et al., 2014). I focus on the intersection of this literature by studying how past experiences with hurricanes affect the responses in the current period. I find that the historical exposure of a area to hurricanes increases the rate at which emergency supplies are sold in a county.

The literature on the salience of disasters tends to use variables with lower purchasing frequencies such as houses and insurance or changes in financial markets to understand how recent disasters affect decisions (McCoy and Walsh, 2018; Bakkensen et al., 2019; Bourdeau-Brien and Kryzanowski, 2020). Others that focus on a more individual response rely on either general surveys or surveys after disasters to understand feelings around risk and preparation steps that had been taken (Lazo et al., 2010; Baker, 2011; Meyer et al., 2014). The closest research to mine is that in the wildfire literature which looks at people's interest in home air purifiers, and the decision to stay indoors (Burke et al., 2022). I add to this literature by utilizing sales data of emergency supplies that are more liquid than a home or insurance and can be purchased much more frequently. Additionally,

emergency supplies are relatively inexpensive, allowing me to capture the salience of hurricanes for a larger portion of the affected population. My findings are similar to those of prior literature in that the largest effect of hurricanes falls off after only a few years.

The rest of the paper proceeds as follows. Section II discusses the background of hurricane preparation. Section III explains the data. Section IV goes over the main results and section V concludes.

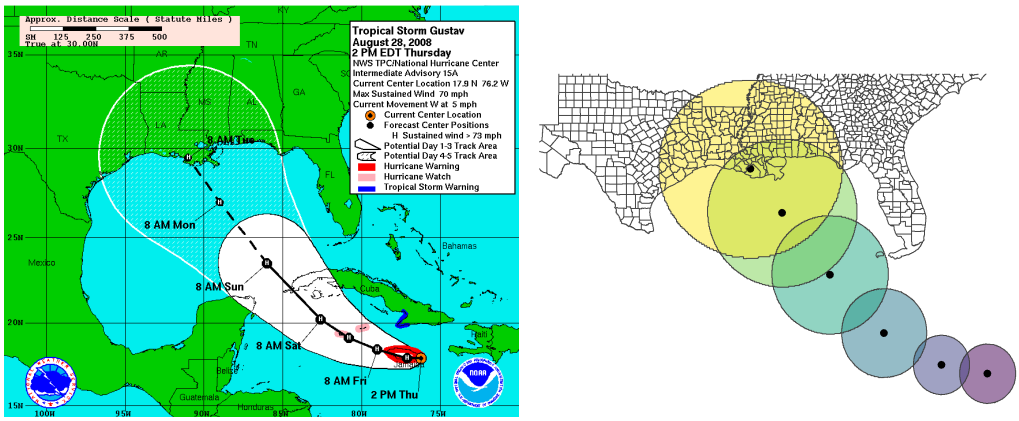
2 Background

The Department of Homeland Security (DHS) along with other federal, state, and local government agencies distribute information on how to prepare for a myriad of natural disasters. Part of preparing for such events is the building a "Disaster Supply Kit." The list of emergency supplies to include in each kit is identical among agencies, as they all use the DHS checklist as a guide. The specifics of this list can be found at the DHS website, ready.gov. This checklist includes items such as purified drinking water, non-perishable foods, batteries, flashlights, first-aid kits, and many more.

When signs of a tropical system (potential hurricane) first appear, the National Oceanic and Atmospheric Administration (NOAA) begins tracking the system. NOAA gathers information on the size, wind speed, movement speed, ocean temperature, and other atmospheric conditions that will affect the tropical system. Using this information, they begin forecasting the system for the next 5 days. Specifically they forecast the expected wind speed and location along with storm surge, tropical storm, and hurricane watches and warnings (NOAA, 2024). If a system's path puts it on a potential collision course with the United States, NOAA increases their forecast frequency to every 6 hours to keep the populous informed about any new developments.

Because of the frequent updates, as well as modern technology providing quick access to information, people have time to respond to the threat of an impending hurricane well before it ever makes landfall. This provides the unique opportunity to observe how the population responds to such information. Additionally, we can observe how prior experiences and knowledge play a role in

Figure 1: Replication of Official NOAA Forecast



Note: Maps of official and replicated forecasts of Hurricane Gustav (2008). Official advisory (left) is from August 28, 2008 at 2pm with the replication (right) made from the archived shape file of the same advisory. Gustav made initial landfall in Louisiana at 10pm on Monday, September 1.

how people unitize the information they are being given in relation to disasters.

3 Data

3.1 Hurricane Data

The National Hurricane Center (NHC) GIS Archive of Tropical Cyclone Forecasts contains all shape files used to build the hurricane forecasts and watch/warning advisories since the 2008 Atlantic Hurricane Season. There is a new file for each advisory update that occurred: every six hours from the system's conception to its dissipation. Additionally, each file contains information on the system's current location and wind speed as well as the forecast paths, wind speed, and weather advisories for the next five days. Using this information, I am able to replicate the forecast as it appeared in the past. Figure 1 shows the results of using the archived shape files to replicate the forecast as it was seen by individuals in the past.

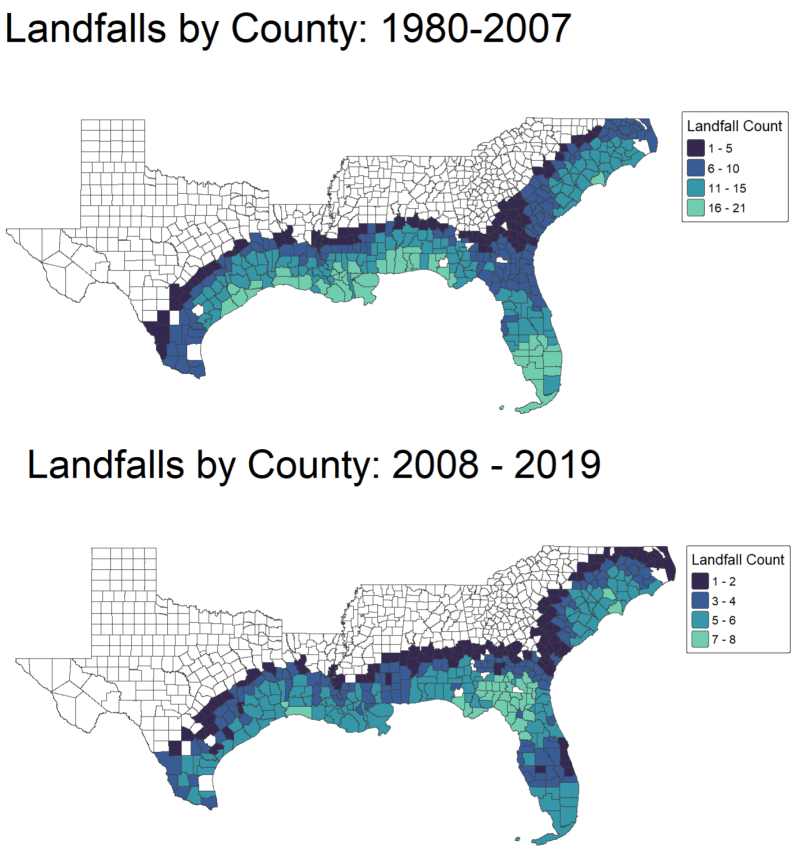
To build out the advisories I follow an approach similar to that of Beatty et al. (2019). By drawing a circle with a 100 nautical mile radius around the current location of the hurricane, I am able to identify any counties that are in the average affected area of any given hurricane. I then draw

circles around the forecast locations in varying sizes. The circle around the one day out location has a radius of 100 nautical miles, two days out has a radius of 150 nautical miles, three days out 200, 4 days out 250, and 5 days out 300. Through this I am able to replicate the forecast track and accompanying cone of uncertainty shown to the public during the initial forecast. This allows me to identify which counties are within the forecast, and at which point they may be affected by the impending hurricane. Because of the speed at which some hurricanes move, occasionally a county is inside of two forecast circles at the same time. While this may seem like a challenge at first glance, it does not pose any issues. Because the movement speed of hurricanes can speed up and slow down with the weather, it is possible for the forecast timing of when a hurricane is predicted to make landfall to shift. Additionally because the advisories are updated every six hours, there are 4 observations for every day. This would result in counties often being in two different forecast circles even without overlap.

In addition to the GIS Archive, I also use NOAA's HURDAT2 (a.k.a. "Best Track") data. This data set contains the location and wind speed of all recorded hurricanes since the early 1900s in six hour increments. Because hurricane tracking did not become consistent until more recently, I only use the data since the 1980s as recommended by NOAA. By combining these two data sets, I observe a complete history of hurricanes since the 1980s. Figure 2 shows the total count of hurricanes to make landfall in each county from 1980-2007, during which I only have the HURDAT2 data, and from 2008-2019, once the GIS Archive is made available. It can be observed from these maps which specific regions of the Southeast are hot-spots for hurricane landfalls. Interestingly, historical landfall counts do not perfectly correlate with the most recent decade of landfall. Specifically with the Gulf Coast from Texas to Alabama seeing fewer hurricanes, while North Florida and Georgia became the most affected by hurricanes.

From combining the HURDAT2 and GIS Archive, I create two new variables to capture different aspects of a county's history with hurricanes. The first variable I create is the total historical landfall count of each county, $HurCount$. This variable will capture the impact of living in an area that is more geographically prone to hurricane landfalls. Additionally, this variable will capture any effects of permanent shifts in hurricane landfall locations due to climate change. To build this variable, I

Figure 2: Total Hurricane Landfalls

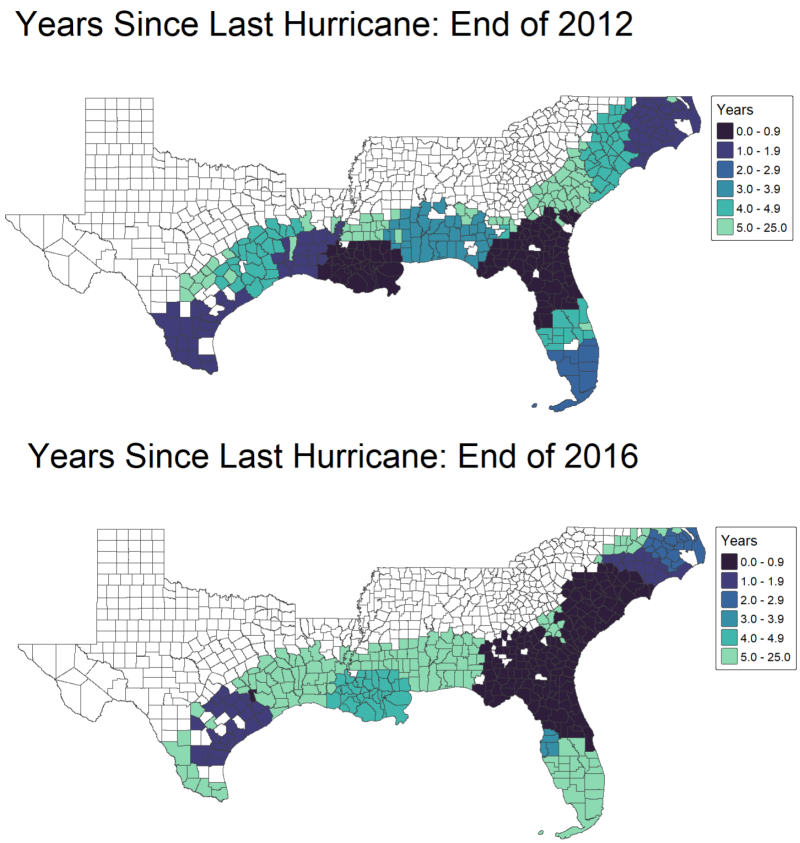


Note: Maps of official total tropical systems to make landfall from 1980-2007(top) and from 2008-2019(bottom). The lighter colors represent more landfalls during the given time frame.

first use the HURDAT2 data to set a baseline for each county. The total number of hurricanes to make landfall in each county from 1980-2007 is used as their starting hurricane count on January 1, 2008. The relative starting values are observed in the top panel of figure 2. At the beginning of each following year, the sum of landfalls in the year prior are then added to the total landfall count.

The second variable I create from this combined data is the length of time that passes between subsequent landfalls, $YrSince$. This variable captures the effects of the cyclical nature of hurricanes season from the El Niño and La Niña cycles. Due to this cycle hurricane season go through

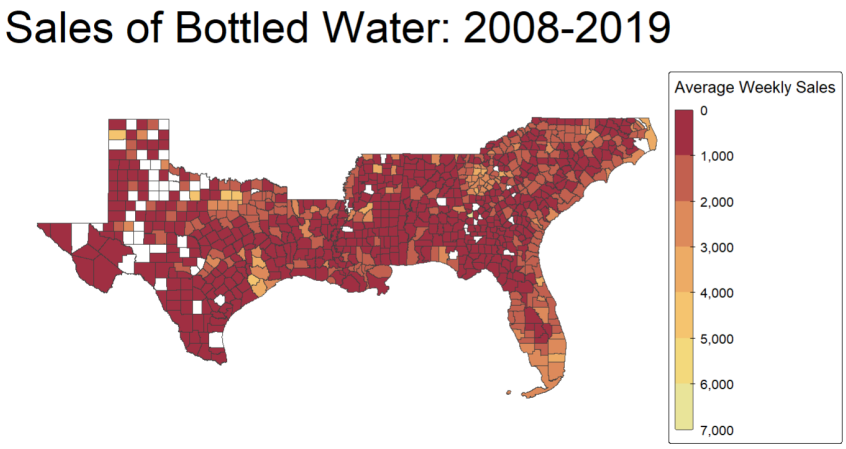
Figure 3: Variation in Timing of Hurricane Landfalls



Note: Maps representing the years that have passed since that last hurricane made landfall in a county. Each map is based on the hurricane record during the last week of the year. Thus, counties with 0 years since landfall in the top map were hit by a hurricane sometime in 2012.

subsequent years of high hurricane counts followed by years of low hurricane counts. This can lead to areas which are more geographically prone to hurricanes to go years without any hurricane activity or vice versa. To build this variable I use the HURDAT2 data to find the hurricane that hit each county most recently to January 1, 2008. Each entry in the data adds another week since the prior hurricane landfall. A week is added to this variable for each entry until the county sees a new landfall. After a county is hit by a new hurricane, the time since the prior storm is reset to account for the new landfall. Figure 3 shows this timing variable at the end of both 2012 and 2016.

Figure 4: Average Sales of Bottled Water



Note: A map of the average total revenue from bottled water sales per store in each county from 2008-2019.

3.2 Scanner Data

Using NielsenIQ Retail Scanner Data, I am able to observe weekly level sales of emergency supplies and other goods across the Southeast. This data contains weekly sales of all goods sold at each reporting store. In addition to the quantity sold, the data contains information on the price, any sales or specials, the sizes of products, and the county that the store is located in. Although the data set covers the entire nation, I choose to limit my analysis to those states most often affected by the Atlantic hurricane season[3]. Additionally, I limit the analysis to only sales of bottled water. I choose bottled water over other products first because it is the first item listed on the official emergency kit checklist. Second, because its low price makes it affordable to the vast majority of consumers. Third, it is a rather homogeneous good that relies less on tastes and preferences than more broad category items like non-perishable foods. Finally, because the scanner data primarily covers grocery type stores, giving much more coverage to water than goods such as flashlights.

In order to understand how the population as a whole is responding to hurricane forecast, I aggregate the data to a county-week level. First I aggregate the sales of individual bottled water products in each store by calculating total revenue from bottled water sales and average price

[3]States included are Alabama, Florida, Georgia, Louisiana, Mississippi, North Carolina, South Carolina, and Texas.

of all water sold. Then I combine the store level sales to the county by calculating the average total revenue per store and the average price per store. I must use the average revenue per store in a county rather than the total revenue per county because only select stores are represented in the data. Without this weighting, some counties would be over or under-represented in their responsiveness to hurricane forecasts. The average weekly total revenue from bottled water sales for each county can been seen in figure 4. This figure shows that Atlanta, Dallas-Forth Worth, Houston, and the Atlantic Coast areas purchase the most bottled water on average.

3.3 Control Data

Because bottled water sales may be affected by high heat days, I use data from NOAA's Global Historical Climatology Network. Average temperature is recorded at the daily level at hundreds of weather stations across each state. I first match each county to its closest weather stations, then I take the average of the temperature across the entire week. While this gets rid of some of the variation, it still provides enough to understand if the week was hotter than average.

There is also significant variation in the population of counties in the data. Due to this, average sales per store are likely to be significantly larger in more populous counties skewing any results. To correct for this, I use population data from the U.S. Census. Using the county level population from the 2010 census, I create a sales per 100,000 residents variable.

4 Methodology

There are two dominant mechanisms for historical experience to play a role in the level of "panic buying" that occurs during a hurricane forecast. The first is the accumulated knowledge of a country through both the passed on knowledge of prior generations and the knowledge gained through accumulated personal experiences. Living in an area that is prone to a certain type of natural disaster allows people to learn about that specific disaster either through personal experiences or through the shared knowledge of those around them, regardless of how long they have lived in the area. As areas are affected by more and more hurricanes, the average knowledge of the population

in those areas may increase following traditional learning models. This would likely reflect in sales of emergency supplies with the total sales during a hurricane forecast approaching an equilibrium amount as the counties observed see more hurricane observations.

The second mechanism likely to influence "panic buying" is the time that has passed since the last hurricane hit an area. Regardless of the baseline knowledge of a county, when a hurricane has just made landfall, the memory of the results is fresh. As time passes memory tends to fade. This memory likely also affects the responsiveness of consumers to hurricane forecasts. This affect is also likely impacted by the strength of the last hurricane as well. If a recent storm was a tropical depression, it would likely not change the area's responsiveness since the damage and impact would be minimal. However, if the last hurricane were a major hurricane, the damage would be extensive, and its impact of the general population much larger. Conditional on the last hurricane strength, an increases in sales as time passes likely indicates that memory has a large affect on "panic buying".

While it may seem that these two variables, overall experience and years between subsequent hurricanes, are collinear in nature and cannot be separately identified, they are actually driven by separate forces. The total experience and therefore the likeliness of a county being hit at any given point in time is purely geographic in nature. South Florida around Miami juts out into the Atlantic Ocean setting it right in the path of hurricanes. However North Florida around Jacksonville, is hardly ever hit due to the currents in the ocean and atmosphere. The relative timing between subsequent events, or the intensity of subsequent hurricane seasons, is far more driven by meteorological conditions, namely the El Niño and La Niña cycle and the Atlantic Multi-Decadal Oscillation (NOAA, 2014). Specifically, being in a La Niña cycle makes weather conditions more favorable for hurricanes to form and remain active, thus increasing the likelihood of all coastal counties to be hit in the year. Vice versa, the El Niño cycle makes conditions unfavorable for hurricanes to form and stay active, thus reducing the chance of any coastal counties getting hit. One clear example of this comes from Miami-Dade County, Florida, which had 15 landfalls from 1980 to 2007, a relatively high number. Miami-Dade was hit by hurricanes in 2008 and 2010, a two year gap, but was not hit again until 2017, a 7 year gap. It was then hit again the same year and the following.

4.1 Event Study Models

I begin my analysis by conducting several event studies to test for more general impacts of county histories with hurricanes. The event studies are conducted using the following regression:

$$Sales_{c,t} = \sum_{\tau=-10,\tau\neq-2}^{10} \beta_{\tau=t} L_{c,\tau=0} + \alpha_c + \alpha_m + \alpha_y + \gamma X'_{c,t} + \varepsilon_{c,t} \tag{1}$$

$Sales_{ct}$ represents the average total revenue per store from sales of bottled water per 100 thousand residents during a given county-week. The variable $L_{c,\tau=0}$ is an indicator variable that is equal to one if a tropical disturbance makes landfall in a county during the event week $\tau = 0$. I use week $\tau = -2$ as the reference period for the event study to avoid spillover from the forecast in week $\tau = -1$. Because the scanner data is reported at a weekly level, if a storm were to make landfall early in the week (i.e. Sunday or Monday), many of the purchases in advance would be done in the sales week before the storm lands. The parameters of interest are $\beta_{\tau=-1}$, $\beta_{\tau=0}$, and $\beta_{\tau=1}$. Positive coefficients on $\tau = -1$ or 0 would indicate that individuals are waiting for a storm to purchase emergency supplies. A null result on these weeks would indicate that most individuals keep a sufficient stock of supplies in case of such an emergency. Average weekly temperature, current hurricane category, and last hurricane category are controlled for in X'_{ct}. County, month, and year fixed effects are represented as α_c, α_m, and α_y respectively. I allow for clustering of the standard errors at the county level.

To test for any effect from changes in total experience or from changes in timing between subsequent storms, I subset the data into quartiles along the $HurCount$ variable and again along the $YrSince$ variable. The quartiles for $HurCount$ are 1 to 7, 8 to 12, 13 to 16, and 17 or more. Because the variable is updated at the beginning of each year, it is possible for counties to move between quartiles or even pass counties that started in higher quartiles. The quartiles for $YrSince$ are less than 1 year, 1 to 2 years, 3 years, and 4 or more years. Again, because the variable is continuously updating, counties may fall into multiple quartiles throughout the panel.

4.2 Interacted Regression Models

After testing for generalized results with event studies, I move to regression analysis to test for the effects of one more hurricane landfall and one more year between subsequent hurricanes, a one unit increase in $HurCount$ and $YrSince$ respectively. The model used in this portion of the analysis follows.

$$\begin{aligned} lnSales_{c,t} = & \beta_1 Threat_{c,t} + \beta_2 Landfall_{c,t} + \\ & \beta_3 H + \beta_4 H \times Threat_{c,t} + \beta_5 H \times Landfall_{c,t} + \\ & \alpha_c + \alpha_m + \alpha_y + \gamma X'_{c,t} + \varepsilon_{c,t} \end{aligned} \tag{2}$$

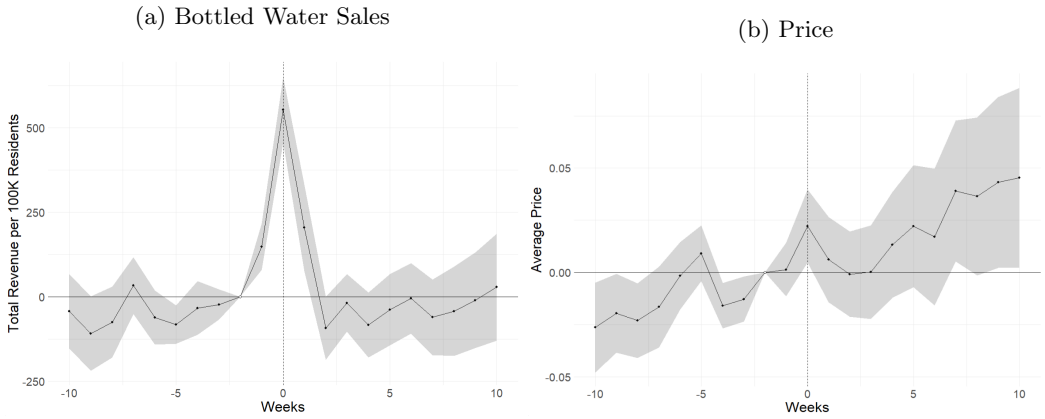
The log of average sales per store for a given county-week is represented by $lnSales_{c,t}$. $Threat$ indicates any current hurricane forecasts, and $Landfall$ indicates any current hurricane landfall. The variable H represents the historic hurricane related variables of interest, $HurCount$ and $YrSince$. Weekly temperature, the current hurricane category, and the prior hurricane category are controlled for in X'. Additionally, there are county level fixed effects, α_c, as well as both year, α_y, and month, α_m to capture any seasonality in sales. The variables of interest are β_4 and β_5 as these will indicate the effects of historical hurricane trends on "panic buying" related sales specifically.

5 Results

5.1 Event Study Results

Before splitting the panel into quartiles, I first test for any aggregate effects of hurricane forecasts and landfalls. By running the event study from equation (1) on all counties, I am able to estimate the average response of all counties to a hurricane. Figure 5 Panel (a) shows a clear sudden increase in sales of bottled water during the weeks surrounding a hurricane landfall with sales increasing by over $500 per one hundred thousand residents in the county. Sales more than two weeks before and after the hurricane are no significantly different from zero indicating the the effect of the hurricane is temporary in nature and not permanent. These results follow closely to those of prior literature,

Figure 5: Aggregate Event Studies of Bottled Water Sales



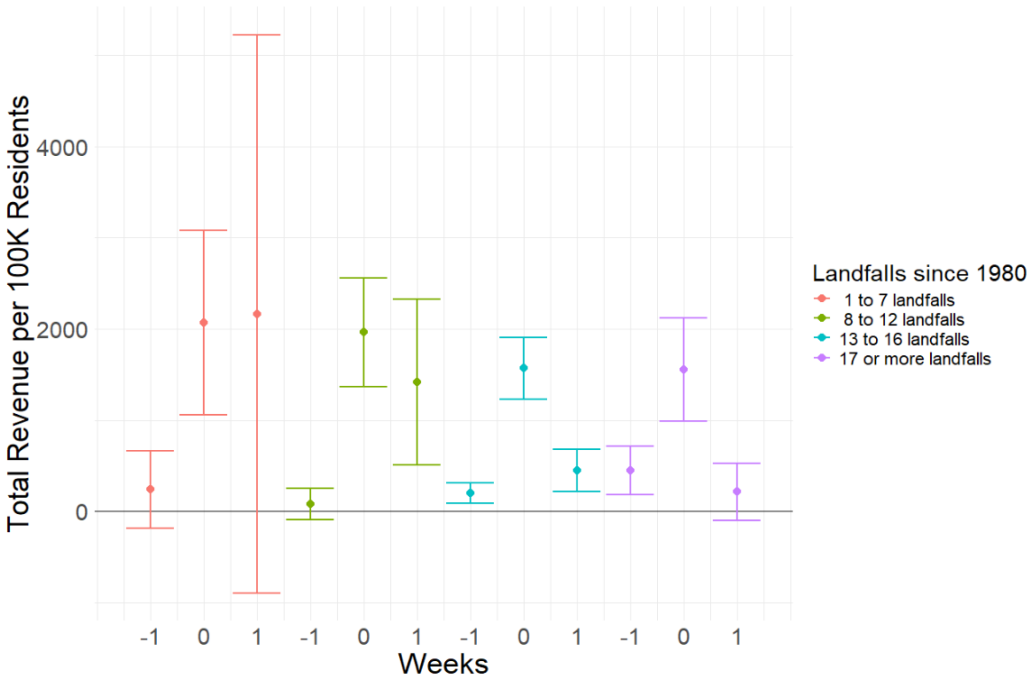
Note: Results from event studies following equation 1. The vertical axis represents total revenue per 100 thousand residents in USD in panel (a) and average weekly price in panel (b). The horizontal axis represents weeks. The reference period is week -2. The vertical dashed line at week 0 represents the week when a storm made landfall. The band around the estimates represents the 95% confidence interval.

namely Beatty et al. (2019).

I also run the event study on the entire panel using price as the outcome variable rather than sales. Since the data is at a weekly level, price is the average price of bottled water in each county for the week. For any changes in sales responses to be behaviorally driven, it must come from a change in the amount that people are purchasing, not from changes in the pricing structure. Figure 5 panel (b) shows the results from the pricing event study. Here there are no significant changes in the price of bottled water, and the largest deviation seen during the hurricane is less than 1 cent. Thus, any changes in sales are likely from changes in consumer behavior.

To see if there are any general effects of a county being located in a more hurricane prone portion of the coast, I rerun the event study above with the panel broken into quartiles along the variable $HurCount$. Figure 6 shows the results of the event study along quartiles. Since the primary times of interest are the weeks immediately surrounding a hurricane landfall, those are the only points shown in the graph. Results of each full event study are show in appendix figure A1. Interestingly, while the hurricane leads to an increase in sales for all counties, there is little difference in sales for counties in different quantiles. The point estimates for the week before the hurricane makes landfall are nearly identical. While there is more variation in the week of landfall, there is not

Figure 6: Event Studies of Historical Exposure on Sales

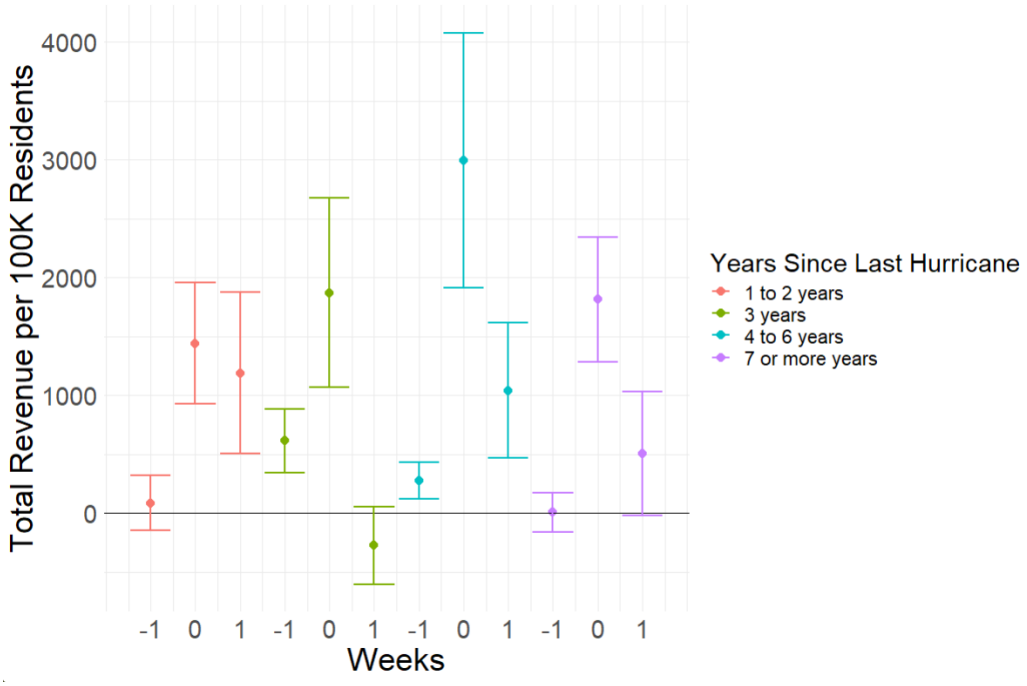


Note: Results from event study with subsets based on historical exposure of the county to hurricanes. The vertical axis represents total revenue per 100 thousand residents in USD. The horizontal axis represents weeks. Week 0 is the week that a tropical disturbance made landfall in a county. The bars represent the 5% confidence interval around the point estimate.

strong evidence that the response differs across groups. Thus, if hurricane exposure affects sales of emergency supplies, it is likely happening through some mechanism other than hurricane landfalls. Two additional methods by which hurricane exposure may affect emergency supply sales is by either changing the average consumption of bottled water in a county, or by changing the sales of bottle water at the beginning of hurricane season. I later test for both of these possible mechanisms.

To test for aggregate effects of repeated high intensity hurricane seasons, I repeat the event study from equation (1) with the panel subset by variable $YrSince$. The results from this analysis are shown in Figure 7. As the years between subsequent hurricane landfalls increases, there is a slight increase in sales during a hurricane for the first five years followed by a sharp decline in hurricane related sales. While the point estimates between each individual group are not significantly different

Figure 7: Event Studies of Timing on Sales



Note: Results from event study with subsets based on the years that have passed since that last storm made landfall in a county. The vertical axis represents total revenue per 100 thousand residents in USD. The horizontal axis represents weeks. Week 0 is the week that a tropical disturbance made landfall in a county. The bars represent the 10% confidence interval around the point estimate.

from one another, the increase in sales from the first quartile to the third is significant, as well as the decline in sales starting at year 6. This increase in sales over the first five years is likely a combination of households having leftover bottled water from previous hurricanes if they are in the first quartile and having less clear memories of the needed quantities as time passes. What is most interesting is the drastic drop in the 6 or more year group. While the mechanism behind this behavior is uncertain, it is possible that as the years between subsequent hurricanes increases, the perceived risk associated with hurricanes is inflated. The increase in bottled water sales is consistent with this notion of risk perception. However, once 6 or more years have passed the perceived risk is so high that the average consumer would rather evacuate leading to the near insignificant change in sales.

5.2 Regression Results

While the event studies above show the impact of having relatively more or less exposure or having relatively more recent exposure to tropical disturbances of different classifications, they do not tell us the effect of a single additional disturbance on sales of emergency supplies in a county. To estimate the effect of an additional historic landfall or an additional year in between subsequent landfalls, I use interacted regression models. I begin by running a model similar to that presented in Beatty et al. (2019) as a baseline. The regression used is as follows

$$lnSales_{c,t} = \beta_1 Threatened_{c,t} + \beta_2 Struck_{c,t} + \beta_3 After_{c,t} + \alpha_c + \alpha_m + \alpha_y + \gamma X'_{c,t} + \varepsilon_{c,t} \quad (3)$$

The outcome of interest is log total revenue from sales of emergency supplies for a given county-week, $lnSales_{c,t}$. $Threatened_{c,t}$ indicates if a county was in the projected path of a storm any day during a given week t. This variable was generated following Beatty et al. (2019): the county must be within the projected path of a storm and the storm is expected to make landfall within the next 72 hours. $Struck_{c,t}$ indicates if the eye of the storm landed within 100 nautical miles of the county. $After_{c,t}$ indicates if a county was struck by a storm in the past 72 hours. This picks up any lagged effect on purchases. The vector of controls, $X_{c,t}$ includes minimum, maximum, and mean temperature for the week and total rainfall in inches. I also include county, month, and year fixed effects and cluster the standard errors two-ways, at the county and year levels.

The results for the regression in equation 3 are presented in Table **??**. I find that when a county is threatened by a storm, bottled water sales increase by 9.7%, and by 15.4% when a county is hit. Battery sales increase by 15.5% and 60.48%, and flashlight sales increase by 41.2% and 160% respectively. These increases in sales follow those seen in the aggregate event study for the same goods. Because the temperature controls are highly correlated with one another, and precipitation is highly correlated with the presence of a storm, I also run the regression omitting all weather controls expect mean temperature. The effect of removing the additional weather controls are minimal and presented in Table **??**. I only use mean temperature as a control variable in regressions for the remainder of the analysis.

Table 1: Base Results

Dependent Variable:	log(total_rev_per_cap)
Model:	(1)
Variables	
Hur_Threat	0.1073
	(0.1564)
Hur_Landfall	0.4481***
	(0.0683)
temp_mean	0.0046***
	(0.0010)
Fixed-effects	
year	Yes
month	Yes
fips	Yes
Fit statistics	
Observations	448,123
R^2	0.77057

Clustered (fips & year) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

To understand the effect of additional tropical disturbance landfall or an additional year since the last landfall, I interact the historical county level characteristics in the following manner.

$$\begin{aligned} lnSales_{c,t} =& \beta_1 Threatened_{c,t} + \beta_2 Struck_{c,t} + \beta_3 After_{c,t} + \beta_4 H_{c,t} \\ &+ \beta_5 Threatened \times H_{c,t} + \beta_6 Struck \times H_{c,t} + \alpha_c + \alpha_m + \alpha_y + \gamma X'_{c,t} + \varepsilon_{c,t} \end{aligned} \tag{4}$$

The variable $H_{c,t}$ captures the historical tropical disturbance characteristic of interest for that regression, historical count or years between landfalls. The historical count variable is created using HURDAT2 data in the following manner. The first week of 2008 is the total amount of storms to hit the county from 1980-2007. Afterwards the value is updated each time the county faces a new storm landfall of the classification of interest. The years between storms are created by determining the date of the last storm of a certain classification to make landfall before week 1 of 2008 for each county and calculating the time since the landfall in years. The value is then updated

Table 2: Historical Exposure Results

Dependent Variable:	log(total_rev_per_cap)	
Model:	(1)	(2)
Variables		
Hur_Threat	0.1073	-0.8372**
	(0.1564)	(0.3462)
Hur_Landfall	0.4481***	1.363***
	(0.0683)	(0.1631)
temp_mean	0.0046***	0.0048***
	(0.0010)	(0.0010)
total_hist_landfall		-0.0831***
		(0.0265)
Hur_Threat × total_hist_landfall		0.0662*
		(0.0319)
Hur_Landfall × total_hist_landfall		-0.0653***
		(0.0121)
Fixed-effects		
year	Yes	Yes
month	Yes	Yes
fips	Yes	Yes
Fit statistics		
Observations	448,123	448,123
R^2	0.77057	0.77291

Clustered (fips & year) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

each week for that county. When the next storm of that classification makes landfall in the county, the date of the most recent storm is updated to reflect the new landfall in the subsequent weeks.

The results in table **??** show the effect of additional hurricane landfalls on sales of emergency supplies in a county. Columns (1) and (2) show that the effect of additional hurricanes making landfall in a county has no significant effect on their sales of bottled water. The response still strongly follows that of the base regression and is driven by the presence of a hurricane threat. Batteries and flashlights sales respond differently to increased hurricane activity as seen in columns (3) through (6). As counties experience more hurricanes, the average sales of flashlights in these counties increases. Additionally, in the week that a hurricane makes landfall in a county, there are

increased sales of both batteries and flashlights as the number of hurricanes that have previously made landfall increases. This indicates that the residents of these counties are learning that they need to have these supplies during a tropical disturbance. Because the sales in these counties are increasing as exposure to hurricanes increases, it is also possible that the need to purchase these supplies is not transparent enough in counties that have not had much historical exposure to hurricanes.

The results for the heterogeneous effects of additional years since the last hurricane are presented in Table **??**. I find that for each additional year since the last hurricane, average sales in a county increase by approximately 1.3% for bottled water and 0.7% for batteries. Sales from bottled water also increases in the week that a tropical disturbance threatens a county conditional on the years since the last hurricane. Additionally, the term $Threatened$ in column (2) has lost all significance. It appears that the number of years since the last hurricane is driving sales of bottled water in counties in the week before a hurricane landfall. These increases likely indicate that individuals have some leftover goods after the storm and are replenishing as time passes. Sales of water during the week before a storm is also consistent with people in a county having less water at home and needed to replenish more as they move farther away in time from the previous hurricane.

Flashlights have a different change in sales from additional years between storms than water and batteries, likely due to their durable nature. As the years since the previous landfall increases, the sales of flashlights actually decline by approximately 1.7% per year. Because flashlights last a long time, individuals do not need to repeat purchases frequently. The increase in sales of 2.68% on the interaction of $YearsBetween$ and $Threatened$ also follows with the durability of the good. The longer it has been since the last hurricane, the more likely a flashlight is to have been broken or misplaced leading to purchases increasing.

The effects of historical count of hurricanes and years between hurricanes when using independent regressions impact different goods at different points in the hurricane preparation process. To see if the results are independent of one another, I run a regression that includes all county hurricane history terms and interactions. The results from this regression are shown in Table **??** and reflect those shown in the prior tables. Sales of bottled water in a county that is threatened

Table 3: Recent Exposure Results

Dependent Variable:	log(total_rev_per_cap)			
Model:	(1)	(2)	(3)	(4)
Variables				
Hur_Threat	0.1073	3.423***	1.156***	1.471**
	(0.1564)	(0.6349)	(0.1942)	(0.5359)
Hur_Landfall	0.4481***	-2.981***	-0.5415**	-0.9158
	(0.0683)	(0.5581)	(0.1980)	(0.5519)
years_since_landfall		0.0147*		
		(0.0071)		
Hur_Threat × years_since_landfall		-0.6599***		
		(0.1870)		
Hur_Landfall × years_since_landfall		0.7993***		
		(0.1256)		
Hur_Threat × yrs_since_sq		0.0223		
		(0.0154)		
Hur_Landfall × yrs_since_sq		-0.0392***		
		(0.0067)		
years_since_landfall_hur			0.0130*	0.0400**
			(0.0072)	(0.0141)
Hur_Threat × years_since_landfall_hur			-0.1390***	-0.2038
			(0.0220)	(0.1447)
Hur_Landfall × years_since_landfall_hur			0.1307***	0.2143
			(0.0199)	(0.1425)
yrs_since_hur_sq				-0.0021**
				(0.0009)
Hur_Threat × yrs_since_hur_sq				0.0031
				(0.0086)
Hur_Landfall × yrs_since_hur_sq				-0.0043
				(0.0083)
Fit statistics				
Observations	448,123	222,245	217,176	217,176
R^2	0.77057	0.78927	0.78932	0.79015

Clustered (fips & year) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

by a storm see significant increases directly related to the length of time that has passed since the last hurricane made landfall. The increases in battery sales in the presence of a storm are primarily driven by the historical exposure of the county to hurricanes. The increases in flashlight sales in the

presence of a storm are largely driven by historical exposure of a county to hurricanes and partially by the length of time that has passed since the last hurricane.

5.3 Robustness

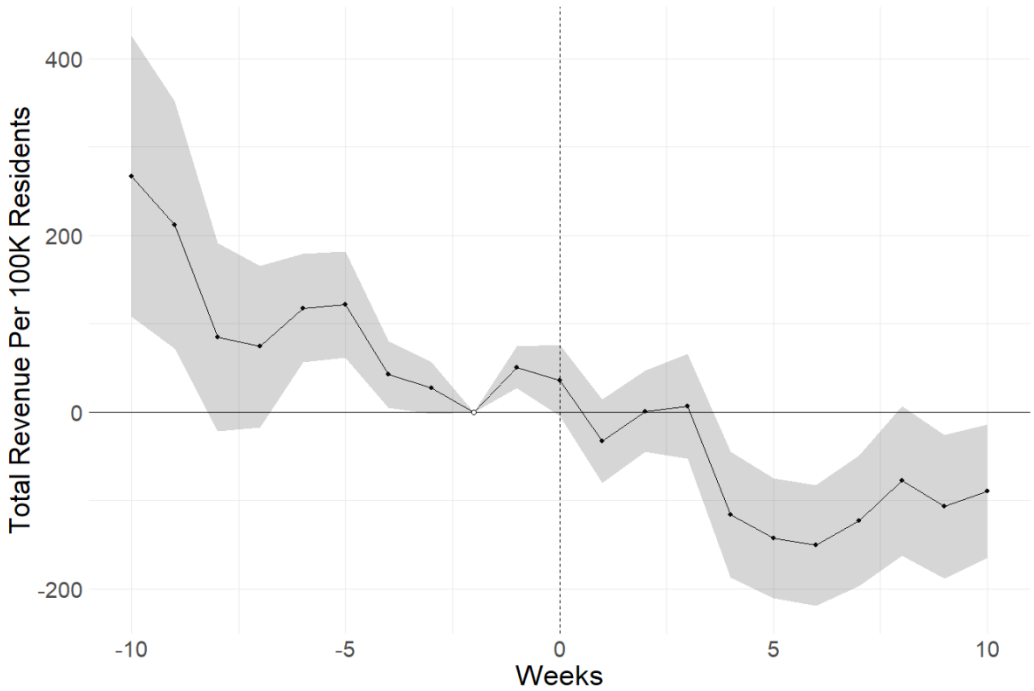
It is possible that the reason for the null response of historical exposure to hurricanes on bottled water sales and the weak response of years between on batteries and flashlights is driven partially by a shift in when people in those counties choose to purchase supplies. Rather than waiting for a hurricane to threaten a county, the average consumer may preemptively purchase emergency supplies at the start of hurricane season. To test for a change in sales at the start of hurricane season, I run a regression similar to the ones above, but with the start of hurricane season as the explanatory variable rather than a hurricane landfall. The official start of hurricane season is June 1 of each year as defined by NOAA. I create an indicator variable for if the week either contains June 1 or is a week immediately before or after June 1 to catch any overlap from the way the scanner data is constructed.

The results for the beginning of hurricane season are in Table **??** of the appendix. The coefficient on the start of season is significant and positive for batteries indicating that regardless of historical characteristics, sales in these counties see a significant increase at the start of summer. Bottled water and flashlights do not see these same increases. None of the coefficients on the interacted terms are significant indicating that the effect of past storms if fully felt by the average response to current storms that was found above.

6 Conclusion

As climate change continues, the intensity of natural disasters and extent of their destruction are expected to increase. Because this destruction also inhibits relief efforts, it is crucial that people know how to prepare for such disasters so that this delay is not harmful. I find that as counties are exposed to more hurricanes, the sales of emergency supplies actually increase in the presence of a storm. This indicates that people in these counties are learning how to prepare for disasters and

Figure 8: Event Studies of Timing on Sales



Note: Results from event study with subsets based on the years that have passed since that last storm made landfall in a county. The vertical axis represents total revenue per 100 thousand residents in USD. The horizontal axis represents weeks. Week 0 is the week that a tropical disturbance made landfall in a county. The bars represent the 10% confidence interval around the point estimate.

adjusting. Additionally, the sales of emergency supplies decline in a way that follows the pattern of risk salience that is seen in other parts of the natural disaster literature when people are faced with a recurring disaster. The combination of needing experience and recent storms to purchase larger quantities of emergency supplies could leave areas with less experience or without a recent storm unprepared when faced with a hurricane. To help reduce the strain on relief efforts during future disasters, it may be useful to find ways to bridge these gaps in emergency purchases.

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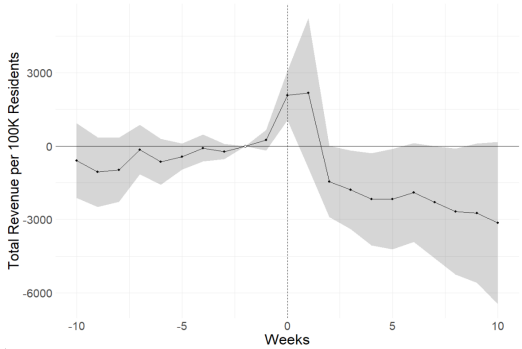
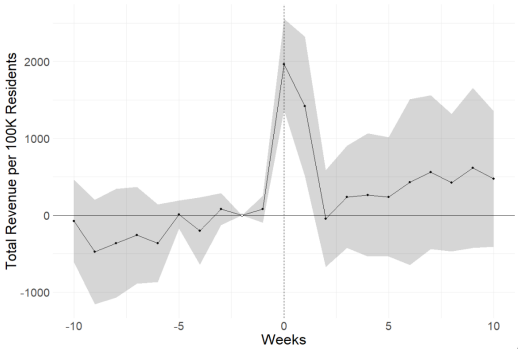
Appendix

Figure A1: Event Studies By Historical Landfalls

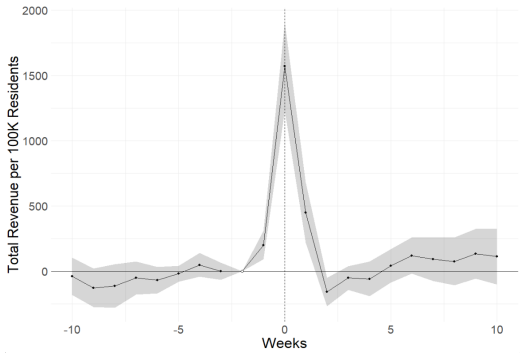
(a) 1 to 7 Landfalls



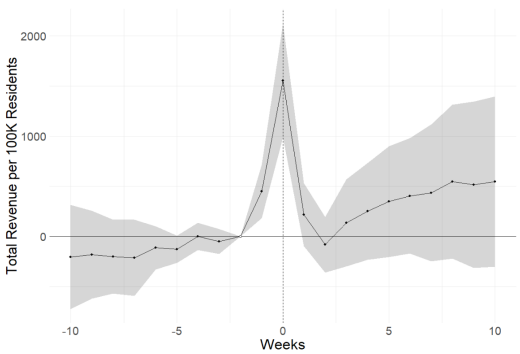
(b) 8 to 12 Landfalls



(c) 13 to 16 Landfalls



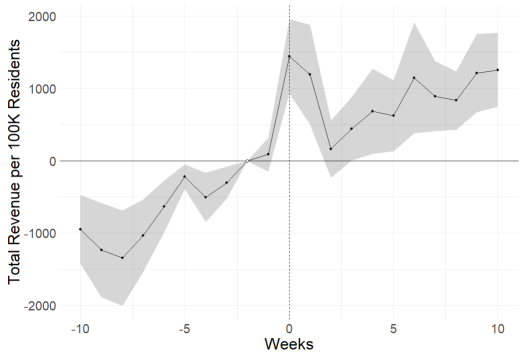
(d) 17 to 25 Landfalls



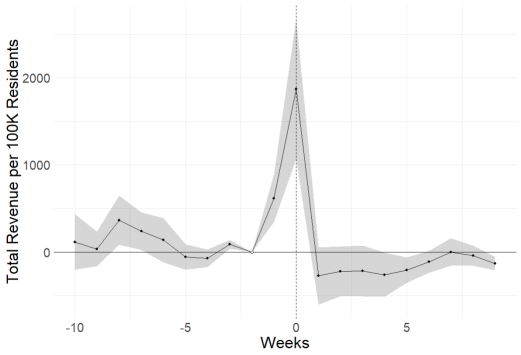
Note: Results from event study with subsets based on the total count of prior hurricanes to hit the county since 1980. The vertical axis represents total revenue per 100 thousand residents in USD. The horizontal axis represents weeks. Week 0 is the week that a tropical disturbance made landfall in a county. The bars represent the 5% confidence interval around the point estimate.

Figure A2: Event Studies By Years Between Subsequent Hurricanes

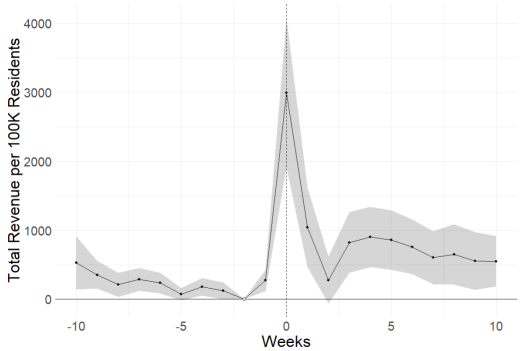
(a) 1 to 2 Years



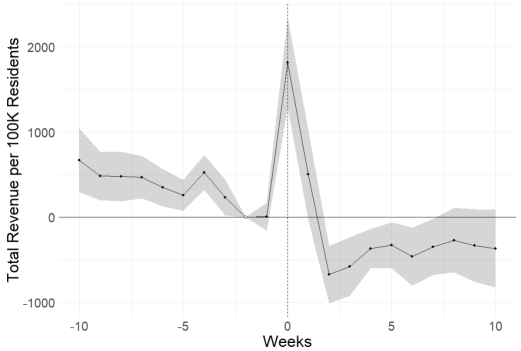
(b) 3 Years



(c) 4 to 6 Years



(d) 7 of More Years



Note: Results from event study with subsets based on the years that have passed since that last storm made landfall in a county. The vertical axis represents total revenue per 100 thousand residents in USD. The horizontal axis represents weeks. Week 0 is the week that a tropical disturbance made landfall in a county. The bars represent the 5% confidence interval around the point estimate.